

Introduction

MAST90083 Computational Statistics and Data Mining

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Outline

§i. Admin

§ii. Introduction & overview

§iii. Basic concepts

Admin

- ▶ Lectures - Associate Professor Karim Seghouane
 - ▶ **Monday:** 15:00-16:00 and **Tuesday:** 13:00-14:00 in PAR-160-G-G01 (JH Michell Theater, 308)
- ▶ Practical lab - Dr. Hangfei Zheng (Subject Assistant)
- ▶ Tutors: Ms Wenjing Yang and Dr. Shangyu Chen
 - ▶ 7 Groups (Nanson Lab, Thompson Lab, and Wilson Lab)
 - ▶ **Monday, Wednesday and Friday:** G6, G5 and G1 from 16:00-17:00.
 - ▶ **Tuesday and Wednesday:** G2 and G4 from 12:00-13:00.
 - ▶ **Wednesday:** G3 from 9:00-10:00 and **Thursday:** G7 from 14:00-15:00.
- ▶ Consultation time
 - ▶ **Monday** from 16:00-17:00 pm, **Tuesday** from 14:00 pn to 15:00 pm and **Wednesday** from 11:00 am to 12:00 pm in office 113 Peter Hall building and via zoom hen requested.

Admin - Assessment

- ▶ Problem solving assignments 40%
 - ▶ Two assignments due mid and late semester
 - ▶ Each written assignment is worth 20%
 - ▶ First assignment will be released on week 5 to be completed and returned on week 7 via LMS
 - ▶ Second assignment will be released on 8 or 9 to be completed and returned on week 10 or 11 via LMS
- ▶ Exam (Hurdle). 60%

Admin - LMS

- ▶ All relevant teaching material will be posted on LMS (including the supplementary and the additional material)
- ▶ Due to the time limits during the lectures, you will need to go over some mathematical details & deepen your knowledge outside of the lectures time.
- ▶ Sessions will be organized to briefly discuss the corrections/responses to the assignment questions and give feedback to students.
- ▶ Semester 2 non-teaching period 29 September to 5 October and SWOT Vac 27 October to 31 October.
- ▶ Last self-enrol date 8 August, census date 1 September, last date to withdraw without fail 26 September and assessment period starts 3 November and ends 21 November.

Admin - General

- ▶ I will be available after the lectures and tutorials for questions and discussion.
- ▶ Attendance to the lectures and tutorials is highly recommended as data shows that students who attend are the students who achieved the best results.
- ▶ Note that lecture capture is an addition for students who attend the lectures, it is not a replacement to attendance.
- ▶ There are three hours of consultation time per week on top of the time after the lectures and tutorials that we will provide.
- ▶ This subject requires 170 hours of total time commitment and at least two hours working on the lecture
- ▶ The library offers a large number of online reading resources

Admin - SSLC

- ▶ SSLC

- ▶ **Data Science Peer to Peer Mentoring Program**

The Data Science Peer to Peer Mentoring Program is aimed towards new Masters and Graduate Diploma in Data Science students, to provide them with an informal space where they can ask questions about their course, future pathways and revise course content with a Mentor, who is a later-year student. Applications are now open for Mentees (students who commenced in sem1 2025)

Note: To be eligible to apply as a Mentor, you must have completed at least one full semester of study.

Admin - References

- ▶ **Elements of Statistical Learning** by Hastie Trevor, Tibshirani Robert & Friedman Jerome (2009).
- ▶ **An Introduction to Statistical Learning** by James Gareth Witten Daniela, Hastie Trevor & Tibshirani Robert (2013).
- ▶ **Computational Statistics** by G.H. Givens & J.A. Hoeting (2005).
- ▶ Academic **articles**, links to **blogs** & **videos** on LMS.

Admin - Communication

- ▶ Note that emails regarding the course material, laboratories and assignments will be addressed during office hours and if time permits.
- ▶ It is expected that questions regarding these matters will be asked during consultation hours or during the laboratories.
- ▶ There will be no consultation hours during non-teaching periods.
- ▶ I will be out of office during the examination period.
- ▶ I will provide extra consultation time on week 12 and during SWOT.
- ▶ Students should plan ahead with any queries regarding assignments or course material.

Lecture schedule (provisional)

- ▶ **Data mining** (7 weeks):
 - ▶ linear regression and regularisation; model selection, kernel and local regression; basis expansion and spline regression; general additive models (GAM); support vector machines (SVM); component analysis and deep learning.
- ▶ **Computational statistics** (5 weeks):
 - ▶ EM algorithm; cross-validation; Monte Carlo Methods and bootstrap methods.

Question to the class...

How would you explain in a few sentences to a general audience

- ▶ What a statistician / data scientist does?
- ▶ What is the main purpose of that?

Some answers

- ▶ What a statistician / data scientist does?
- ▶ **Creat a model (box) to understand the relationship between several variable**
- ▶ **make sense of data, extract important patterns and trends, we can call this learning from data**
- ▶ What is the main purpose of that?
- ▶ **Predict/decide or describe/understand**

The power of (statistical) model



Sample



Population

Once upon a time ...

Having the data → holding the power !!!



Image source: <http://freshlearners.blogspot.com.au/2015/07/most-recent-communications-technology.html>

Today

The real power is in **KNOWING WHAT TO DO** with the data

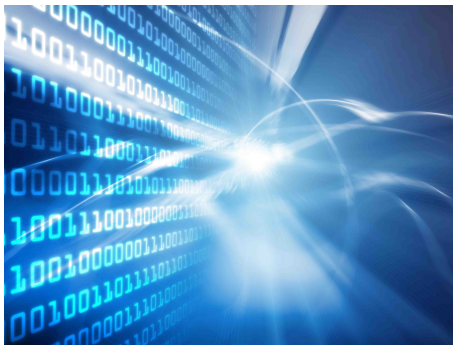
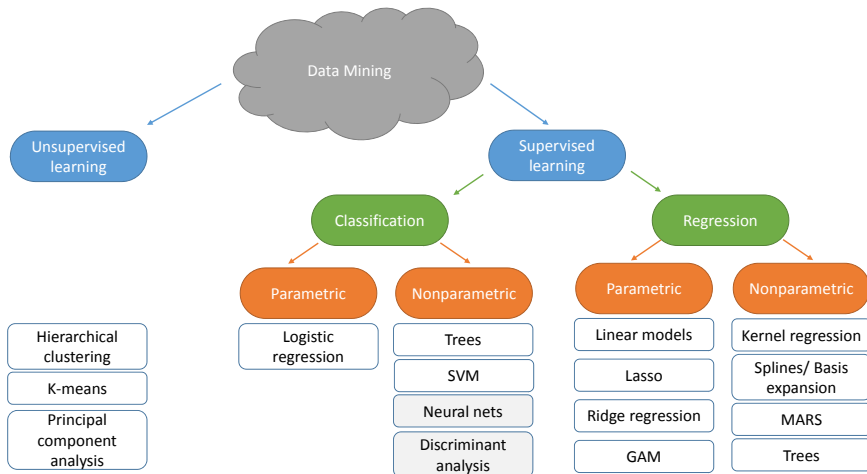


Image source: <http://psmit.com/about.html>

Taxonomy of covered methods



Regression vs. Clasification

- ▶ **Regression** - The aim is to predict the numerical outcome of a subject based on its features.
- ▶ **Clasification** - The aim is to predict the class belonging of a subject based on its features.

Parametric vs. Nonparametric

- ▶ **Parametric** approach - Makes an explicit assumption about the functional form. (Restriction on the shape)
- ▶ **Nonparametric** approach - does not assume any functional form for the underlying model structure.

Parametric models

Advantages

- Simpler to understand and interpret
- Fast to fit (CPU time)
- Requires to fit a small number of parameters, therefore requires relatively small data set

Disadvantages

- Highly constrained to the specified function form.
- It is hard to match the mathematical form of the model to the unknown 'DGP'.
- Can fit well relatively simple data structures (complex models -> danger of over-fitting).

Nonparametric models

Advantages

- Very flexible
- Weak(er) assumptions about the underlying function.
- Outperforms in prediction context, mostly.

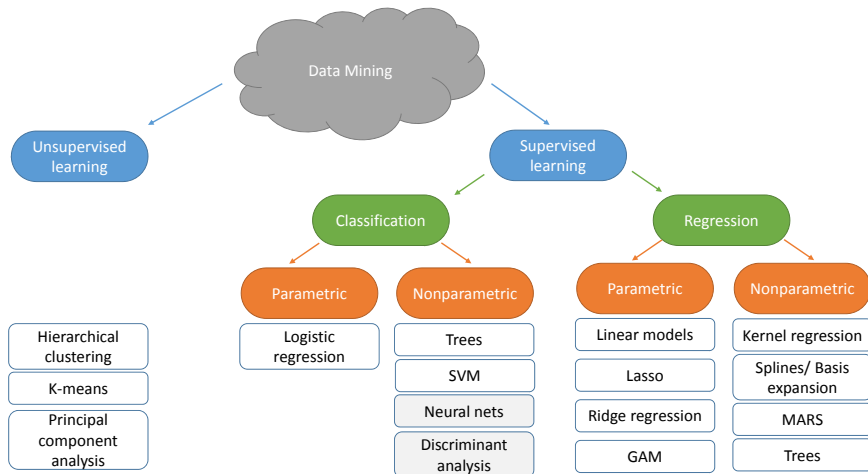
Disadvantages

- Slow(er) to fit (CPU time).
- Requires large(r) data sets
- Harder to interpret
- Higher danger of over-fitting

Supervised vs. Unsupervised

- ▶ **Supervised** learning - In case our data set contains the response (outcome) measurements, the fitted model relates the different features to the response.
- ▶ **Unsupervised** learning - In case our data set contains only information about the different features of the subjects, the fitted model aims to segment the subjects into groups or just learn about the relationship of the features.

Taxonomy of covered methods



How models are fitted

- ▶ By optimization
- ▶ We minimize some criterion (e.g. squared error with some additional penalty)

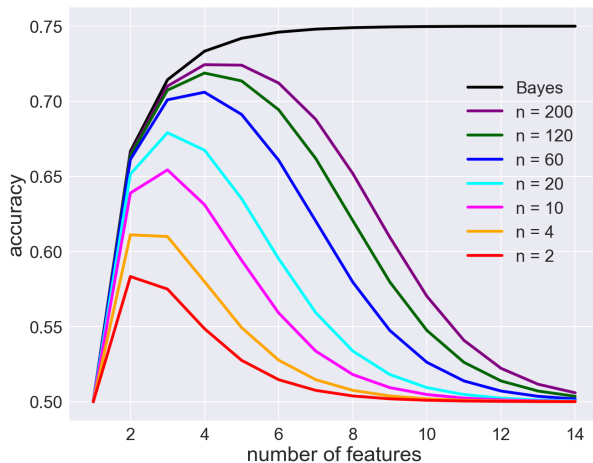
Trade-offs

- ▶ Flexibility (predictability) - Interpretability trade-off
- ▶ Bias - Variance trade-off.
 - ▶ variance - how much $\hat{f}$ changes with different data set.
 - ▶ bias - the error from the approximation.

Complexity

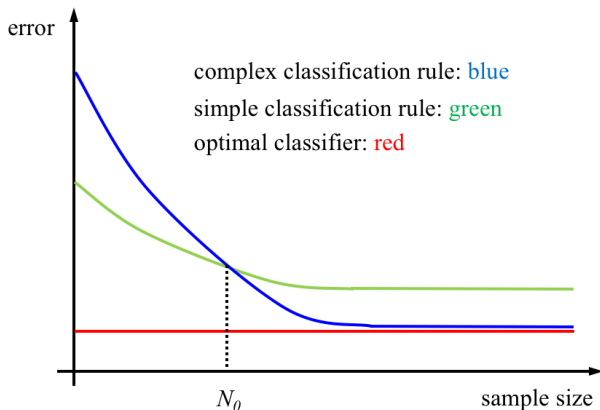
- ▶ Complexity trade-offs involving sample size, dimensionality and empirical performance. It is a characteristic feature of supervised learning methods.
- ▶ Notion of *curse of dimensionality*: for a fixed sample size, the expected classification error will improve by increasing the number of features, but eventually will decrease. This is a consequence of the large size of high-dimensional spaces, which require correspondingly large training sample sizes.
- ▶ *Scissors effect*: the expected error typically decreases as sample size increases, and more complex classification rules achieve smaller error for large sample sizes; however, simpler classification rules can perform better under small sample sizes, by virtue of needing less data.

Complexity



Expected accuracy in a discrete classification problem for various training sample sizes as a function of the number of predictors.

Complexity



Expected error as a function of sample size for two classification rules. There is a problem-dependent critical sample size N_0 , under which one should use the simpler classification rule.